Sahaj Raj Malla

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Education

B.Sc. in Computational Mathematics Dec 2022 – Jul 2027 (expected)
Kathmandu University, Dhulikhel, Nepal

Selected Training & Fellowships

Aspire Leaders Program, Aspire Institute (Harvard University)	Jan 2025 – Mar 2025
Annual Nepal AI School (2023, 2024 cohorts), NAAMII	May 2023, Dec 2024
Womanium Quantum 2023, Womanium & QWorld	Jul 2023 – Aug 2023
AI Fellowship 2023, Fusemachines Nepal	Jan 2023 – Jan 2024
Data Science Fellowship with DataCamp, Code for Nepal	Dec 2021 – Aug 2023
Introduction to Quantum Computing, The Coding School & IBM	Sep 2021 – Apr 2022

Publications

Peer-Reviewed Publications

1. **S. R. Malla**. **MallaNet residual branch merge convolutional neural network with homogeneous filter capsules for Devanagari character recognition.** *Scientific Reports (Springer Nature), 16, 1175*, Dec 2025.
   99.71% test accuracy on the Devanagari Handwritten Character Dataset, 56.41% fewer parameters than prior SOTA; introduces optimized residual blocks and Homogeneous Filter Capsule layers for multiscale feature extraction.
   [**Single Author | Q1 Journal**]
2. **S. R. Malla**. **Heuristic optimization of AI agents for Dhumbal.** *Scientific Reports (Springer Nature)*, Sep 2026.
   Compared AI strategies for the Dhumbal card game; an aggressive heuristic agent reached an 88.9% win rate.
   [**Single Author | Q1 Journal**]
3. **S. R. Malla**. A landmark-based addressing framework for urban navigation using geospatial clustering and pathfinding algorithm. *Kathmandu University Journal of Science, Engineering and Technology, 19(1)*, Feb 2025.
   37.7% lower average path length and 22.9% lower travel time vs. current systems.
4. B. Kafle, **S. R. Malla**, A. Shrestha, S. Jaiswal. Evaluating XGBoost Algorithm for Nepali Stock Price Prediction. *Int. Conf. on Technologies for Computer, Electrical, Electronics & Communication (IEEE)*, Oct 2023.
   XGBoost regression over 241,023 daily observations for 204 NEPSE-listed companies (2013–2023).

Preprints & Manuscripts Under Review

5. **S. R. Malla**. ArthaVyu: A Stacking Ensemble Framework for Multi-Source Real Estate Price Prediction in Nepal. *Research Square*, Jun 2026.
   Stacking ensemble framework for multi-source real estate price prediction in Nepal; integrates property listings, geospatial data, and market indicators for improved valuation accuracy.
6. **S. R. Malla**. Kalimati Vegetable Price Index Forecasting with a Momentum Corrected Online Stacking Ensemble. *arXiv:2605.30720*, May 2026.
   Built the KVPI, an inverse-volatility-weighted index over 135 daily wholesale commodities (2013–2023); Proposed ensemble reached RMSE 1.771, MAPE 0.68% at 90 days, explaining 84.5% of variance.
7. **S. R. Malla**. External Demand, Domestic Monetary Conditions, and Remittance Dynamics in Nepal. *arXiv:2605.19401*, May 2026.
   ARDL-ECM model of remittances as a share of Nepal's GDP (1993–2024) using PCA-derived monetary indices.
8. **S. R. Malla**, S. Kayastha, R. Suwal, H. C. Bhandari, R. Adhikari. XGBoost Forecasting of NEPSE Index Log Returns with Walk-Forward Validation. *arXiv:2601.08896*, Jan 2026.
   Walk-forward validation framework for time-series forecasting of NEPSE index log returns with XGBoost.
9. P. Panth, **S. R. Malla**. Secure, Verifiable, and Scalable Multi-Client Data Sharing via Consensus-Based Privacy-Preserving Data Distribution. *arXiv:2601.00418*, Jan 2026.
   Consensus-driven privacy-preserving framework for secure, verifiable multi-client data sharing.
10. **S. R. Malla**. Devanagari Digit Recognition using Quantum Machine Learning: A Hybrid Quantum-Classical Approach. *arXiv:2506.09069*, Jun 2025.
    First hybrid quantum-classical architecture for Devanagari digit recognition (CNN + 10-qubit variational circuit); 99.80% test accuracy, 0.9980 average per-class F1.

Professional Experience

Founder Nov 2023 – Present
HamroNiwas, Banepa, Nepal

- Built a rental & marketplace platform for Nepali students, connecting 4,000+ users and reducing search time by 90%.

Data Consultant (Contract) May 2025 – Jun 2025
YoungInnovations Pvt. Ltd., Kathmandu, Nepal

- Built ELT pipelines with Apache Spark for large-scale dataset processing; automated workflows with Apache Airflow.

Data Scientist Sep 2021 – Jan 2023
Naamche Technology, Kathmandu, Nepal

- Scraped & analyzed 1M+ real estate listings (Florida, Texas, California); produced investment-decision reports, speeding report computation roughly threefold.

Data Engineer Mar 2021 – Sep 2021
Devfinity Nepal, Kathmandu, Nepal

- Built a Django-based data factory processing 1M+ job records daily, deployed on Azure.

Teaching & Volunteering

Teaching Assistant, Annual Nepal AI School 2024 Dec 2024 – Jan 2025
NAAMII, Kathmandu, Nepal

Website Manager & Executive Member May 2023 – May 2024
Kathmandu University Robotics Club, Dhulikhel, Nepal

Lead Quantum Computing Researcher Mar 2021 – Mar 2023
OneQuantum, Munich (Remote)

Data Scientist & Software Developer Mar 2021 – Mar 2022
Robotics Association of Nepal, Kathmandu, Nepal

Conference Talks & Panels

Conference Presentation, Nepali Stock Price Prediction Oct 2023
IEEE, Bhaktapur, Nepal

Conference Presentation, Landmark-Based Addressing Oct 2024
Kathmandu University, Dhulikhel, Nepal

Panelist, National Youth Scientific Conference 2024 Jun 2024
National Youth Council Nepal, Kathmandu, Nepal

Guest Speaker, Career in AI/ML Dec 2023
CSIT Association of Nepal, Pokhara, Nepal

Selected Projects

DistilBERT, Nepali News Classification Aug 2022
Associated with IT MEET 2022
DistilBERT sequence-classification model over 9 Nepali newsgroup datasets; 95.475% accuracy.

GreatAPI Jun 2022 – Present
Personal project, open-source Python web framework
Full-stack FastAPI framework for AI; self-built and maintained.

SahajMails Nov 2025 – Present
Personal project, open-source Python library
AI Personalized bulk email tool for outreach, newsletters, and campaigns; secure, cost-free email automation.

Honors & Awards

Winner, Infinity Codewave 2026 Hackathon, Kathmandu University — May 2026
Winner, Infinity Data Visualization Hackathon, Kathmandu University — May 2026
Winner, Computer Engineering Model Demonstration, Kantipur Engineering College — Dec 2024
Winner, KU Hackfest 2023 (OSM Category), Kathmandu University Computer Club — Oct 2023

Winner, Seeds for the Future Nepal, Huawei Technologies Co., Ltd. Sep 2023
Winner, Infinity Codewave, Department of Mathematics, Kathmandu University Aug 2023
Winner, AI Competition (Text Challenge), IT MEET 2022, Kathmandu University Aug 2022
Winner, AI Competition (Data Visualization), IT MEET 2022, Kathmandu University Aug 2022
Winner, Youth Innovation Challenge 2022, Social Changemakers and Innovators Aug 2022

Press & Media

Making Devanagari Handwriting Accessible: The Story of MallaNet, *Business 360°* Feb 2026
AI Model to Recognize Handwritten Devanagari Script, *Techpana* (in Nepali) Dec 2025
AI Agent for the Dhumbal Card Game, *Techpana* (in Nepali) Sep 2026
South Asia's 1st Quantum AI For Devanagari Digits – 99.8% Accuracy Breakthrough, *ICT Frame* Jun 2025
Paths of Possibility Shown by Sahaj, *Terai Online* (in Nepali) Apr 2025
A Journey with "Seeds for the Future 2023", *ICT Frame* Nov 2023

Technical Skills

Deep Learning: PyTorch; CNN, residual, and capsule architectures; parameter-efficient model design; transformers
Agents & RL: LangGraph; agent simulation and evaluation; game environments and strategy benchmarking
Quantum ML: Qiskit, PennyLane; hybrid quantum-classical models
Forecasting & Data: XGBoost, stacking ensembles; Apache Spark, Apache Airflow
Engineering: Python, C++, R, TypeScript; Docker, AWS, Azure; Django, FastAPI
Mathematics: Optimization, numerical methods, graph theory, probability & statistics